INCH-POUND

MS51521B
17 March 2016
SUPERSEDING
MS51521A
17 January 1979

DETAIL SPECIFICATION SHEET

ELBOW, TUBE, 90 DEGREE, SWIVEL, 37 DEGREE FLARED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Inactive for new design after 17 August 1999. For new design, use SAE-J514.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18866.

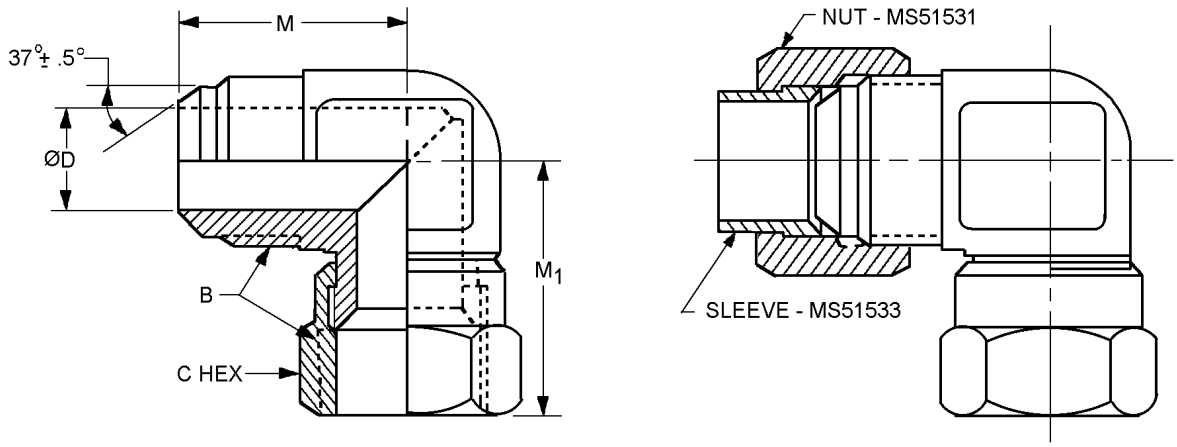


FIGURE 1. 90° elbow tube.

AMSC N/A

FSC 4730

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Dash designator		Tube OD nom.	B Straight thread Class A or B	C Hex flat Nom.	D diameter	
Assembly	Body (see note 7)				Basic inches (mm)	Tolerance inches (mm)
A2	B2	.1250 (3.175)	.3125-24 UNF-2	7/16	.062 (1.57)	±.003 (0.08)
A3	B3	.1875 (4.763)	.3750-24 UNF-2	1/2	.125 (3.18)	
A4	B4	.2500 (6.350)	.4375-20 UNF-2	9/16	.172 (4.37)	
A5	B5	.3125 (7.936)	.5000-20 UNF-2	5/8	.234 (5.94)	
A6	B6	.3750 (9.525)	.5625-18 UNF-2	11/16	.297 (7.54)	±.004 (0.10)
A8	B8	.5000 (12.700)	.7500-16 UNF-2	7/8	.391 (9.93)	
A10	B10	.6250 (15.875)	.8750-14 UNF-2	1	.484 (12.29)	
A12	B12	.7500 (19.050)	1.0625-12 UN-2	1 1/4	.609 (15.47)	±.005 (0.13)
A14	B14	.8750 (22.225)	1.1875-12 UN-2	1 3/8	.718 (18.24)	
A16	B16	1.0000 (25.400)	1.3125-12 UN-2	1 1/2	.844 (21.44)	±.007 (0.18)
A20	B20	1.2500 (31.750)	1.6250-12 UN-2	2	1.078 (27.38)	+.008 -.005 (+0.20 -0.13)
A24	B24	1.5000 (38.100)	1.8750-12 UN-2	2 1/4	1.312 (33.32)	
A32	B32	2.0000 (50.800)	2.500-12 UN-2	2 7/8	1.781 (45.24)	+.010 -.005 (+0.25 -.013)

Dash designator		M inches (mm) ±.030 (0.08)	M_1 inches (mm) ±.060 (0.15)
Assembly	Body		
A2	B2	.770 (19.56)	.970 (24.64)
A3	B3	.830 (21.08)	1.000 (25.40)
A4	B4	.890 (22.61)	1.000 (25.40)
A5	B5	.950 (24.13)	1.060 (26.92)
A6	B6	1.060 (26.92)	1.250 (31.75)
A8	B8	1.250 (31.75)	1.380 (35.05)
A10	B10	1.450 (36.83)	1.620 (41.15)
A12	B12	1.660 (42.16)	1.750 (44.45)
A14	B14	1.730 (43.94)	1.780 (45.21)
A16	B16	1.810 (45.97)	2.000 (50.80)
A20	B20	2.060 (52.32)	2.310 (58.67)
A24	B24	2.330 (59.18)	2.590 (65.79)
A32	B32	3.060 (77.72)	3.380 (85.85)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Break all sharp edges and remove all burrs and slivers.
4. Dimensions and tolerances not shown shall be in accordance with SAE-J514 for 37° flared fittings.
5. The drawing is for identification purposes only and is not intended to restrict designs and shapes not dimensioned.
6. Threads shall be in accordance with ASME B1.1.
7. Bodies will be used for production only and shall not be stocked for maintenance.

FIGURE 1. 90° elbow tube - Continued.

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REQUIREMENTS:

Fittings shall be as specified on figure 1 and in tables I and II.

Components of the assemblies shall be of the same material and finish.

Materials shall be in accordance with MIL-DTL-18866 and table I.

TABLE I. Materials.

Material	Form	Specification	Alloy
Carbon steel	Bar	SAE-J403	1117, 1126, 1140, 1213, 1215, or 12L14
	Forgings		
Chrome-molybdenum steel	Bars	SAE-AMS6370	4130
	Forgings	SAE-AMS6382	
		SAE-AMS6370	
Corrosion resistant steel	Bars and forgings	ASTM A276/A276M	304, 304L, 316, or 321
		ASTM A564/A564M	XM-12 (15-5 PH) UNS S15500 or 630 (17-4 PH) UNS S17400
		SAE-AMS5639	UNS S30400
		SAE-AMS5645	UNS S32100
		SAE-AMS5647	UNS S30403
		SAE-AMS5743	UNS S35500
	Bar	ASTM A582/A582M	UNS S30300
Nickel-copper alloy	Bar	ASTM B164 QQ-N-281	UNS N04400
High-chromium nickel alloy	Bar	ASTM B166	UNS N06690
	Forgings	ASTM B564	
Titanium 1/	Bars	SAE-AMS4928	6AI-4V annealed
	Forgings		

1/ Titanium shall not be used in oxygen or potable water systems.

Finish. Finishes shall be as specified in table II. All platings shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The fittings shall show no evidence of corrosion after 96 hours of salt spray. Fluid passages, other openings and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

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TABLE II. Material and finish identification codes.

PIN code material/plating finish	Material	Plating finish
Blank	Steel	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 2 or SAE-AMS-QQ-P-416, type II, class 2. 1/
H	Steel	Zinc-Aluminum in accordance with ASTM F1136/F1136M, grade 3, NC.
J	Steel	Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 3/
M	Nickel-copper alloy UNS N04400	No additional finish.
N	High-chromium nickel alloy UNS N06690	No additional finish.
P	Steel	Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class 4. 2/
R	Steel	Zinc plating in accordance with ASTM B633; type VI, Fe/Zn 12. 5/
S	Corrosion resistant steel	No additional finish. Passivation in accordance with SAE-AMS2700, method 1, type 6 or 7.
T	Titanium	Anodize in accordance with SAE-AMS2488 type 2. 4/
TF	Titanium	Fluoride phosphate in accordance with SAE-AMS2486. 4/
Z	Steel	Zinc plating in accordance with ASTM B633; type II or III, Fe/Zn 12, or ASTM B695, type II, class 12. 5/
ZC	Steel	Zinc may be any zinc plating from PIN codes H, J, and R with a colored chromate coating 5/

1/ Embrittlement test need not be run. Cadmium shall not be used in oxygen or potable water systems.
2/ Hexavalent chromium free. Finish shall be ROHS compliant.
3/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum coating thickness.
4/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to FED-STD-595 colors 36076 through 36293.
5/ Not for use in aircraft. Requires approval from the Program Officer for all applications.

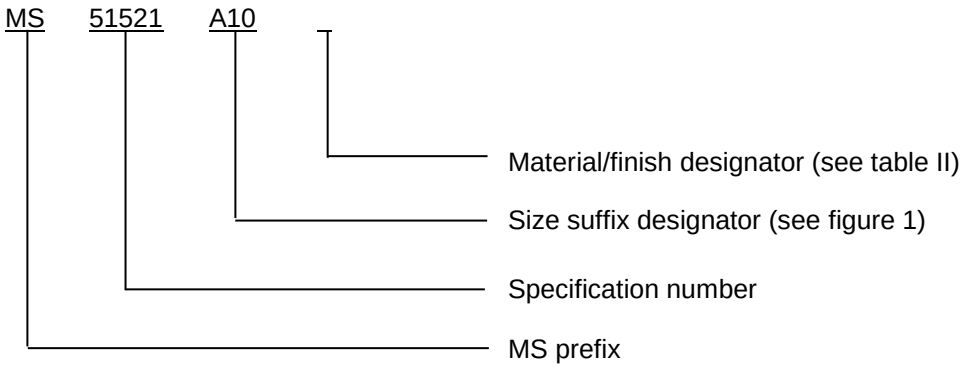
Trivalent wrenchability. When the finish has been damaged due to poor wrenchability, the surface of the connector shall be touched up using one of the brush plating processes below as appropriate to primary finish. The term “trivalent wrenchability” is used to evaluate the ability of the finish to withstand abrasion from an excessive amount of wrenching

a. Brush plating of hard chromium by electrodeposition shall be in accordance with SAE-AMS-2451/5.
b. Brush plating of medium-hardness, low stress nickel by electrodeposition shall be in accordance with SAE-AMS-2451/9.

Maximum operating pressure. Maximum operating pressure shall be in accordance with SAE-J514.

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PIN: The PIN consists of the letters “MS”, the specification number, a letter and number for elbow size, and a letter for material finish designator.



PIN example: MS51521A10 indicates a 90° elbow, tube, .6250 inch (15.875 mm), steel with cadmium plating.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Table III provides a detailed cross-reference of inactive MS51521 PIN’s and for new design SAE-J514 PIN’s.

MS51521 parts have straight threads in accordance with ASME B1.1 the SAE parts have straight threads in accordance with SAE-J425.

Plating “P” SAE allows a range of nickel from 6% to 20%. Below 12%, ZnNi is not much better than zinc plating, which is less expensive and easier to apply. Above 16%, ZnNi becomes more cathodic and no longer acts as a sacrificial coating - if a high nickel coating is damaged the steel beneath the coating will corrode at an accelerated rate.

Users are cautioned to evaluate replacement parts for their particular application.

CAUTION: The superseding information is valid as of the date of this specification and may be superseded by subsequent revisions of the superseding document.

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TABLE III. MS51521 to SAE-J514 cross reference.

Inactive for new design MS51518- PIN		TUBE O.D.	For new design SAE-J514 PIN	
Assembly	Body		Assembly	Body
MS51521A2	MS51521B2	1/8	Not available	Not available
MS51521A2H	MS51521B2H	1/8	Not available	Not available
MS51521A2J	MS51521B2J	1/8	Not available	Not available
MS51521A2M	MS51521B2M	1/8	J514-2-2-070221MA	J514-2-2-070221MB
MS51521A2N	MS51521B2N	1/8	J514-2-2-070221NA	J514-2-2-070221NB
MS51521A2P	MS51521B2P	1/8	J514-2-2-070221PA	J514-2-2-070221PB
MS51521A2R	MS51521B2R	1/8	Not available	Not available
MS51521A2S	MS51521B2S	1/8	J514-2-2-070221SA	J514-2-2-070221SB
MS51521A2T	MS51521B2T	1/8	J514-2-2-070221TA	J514-2-2-070221TB
MS51521A2TF	MS51521B2TF	1/8	Not available	Not available
MS51521A2Z	MS51521B2Z	1/8	J514-2-2-070221ZA	J514-2-2-070221ZB
MS51521A2ZC	MS51521B2ZC	1/8	Not available	Not available
MS51521A3	MS51521B3	3/16	Not available	Not available
MS51521A3H	MS51521B3H	3/16	Not available	Not available
MS51521A3J	MS51521B3J	3/16	Not available	Not available
MS51521A3M	MS51521B3M	3/16	J514-3-3-070221MA	J514-3-3-070221MB
MS51521A3N	MS51521B3N	3/16	J514-3-3-070221NA	J514-3-3-070221NB
MS51521A3P	MS51521B3P	3/16	J514-3-3-070221PA	J514-3-3-070221PB
MS51521A3R	MS51521B3R	3/16	Not available	Not available
MS51521A3S	MS51521B3S	3/16	J514-3-3-070221SA	J514-3-3-070221SB
MS51521A3T	MS51521B3T	3/16	J514-3-3-070221TA	J514-3-3-070221TB
MS51521A3TF	MS51521B3TF	3/16	Not available	Not available
MS51521A3Z	MS51521B3Z	3/16	J514-3-3-070221ZA	J514-3-3-070221ZB
MS51521A3ZC	MS51521B3ZC	3/16	Not available	Not available
MS51521A4	MS51521B4	1/4	Not available	Not available
MS51521A4H	MS51521B4H	1/4	Not available	Not available
MS51521A4J	MS51521B4J	1/4	Not available	Not available
MS51521A4M	MS51521B4M	1/4	J514-4-4-070221MA	J514-4-4-070221MB
MS51521A4N	MS51521B4N	1/4	J514-4-4-070221NA	J514-4-4-070221NB
MS51521A4P	MS51521B4P	1/4	J514-4-4-070221PA	J514-4-4-070221PB
MS51521A4R	MS51521B4R	1/4	Not available	Not available
MS51521A4S	MS51521B4S	1/4	J514-4-4-070221SA	J514-4-4-070221SB
MS51521A4T	MS51521B4T	1/4	J514-4-4-070221TA	J514-4-4-070221TB
MS51521A4TF	MS51521B4TF	1/4	Not available	Not available
MS51521A4Z	MS51521B4Z	1/4	J514-4-4-070221ZA	J514-4-4-070221ZB
MS51521A4ZC	MS51521B4ZC	1/4	Not available	Not available

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TABLE III. MS51521 to SAE-J514 cross reference - Continued.

Inactive for new design MS51518- PIN		Tube O.D.	For new design SAE-J514 PIN	
Assembly	Body		Assembly	Body
MS51521A5	MS51521B5	5/16	Not available	Not available
MS51521A5H	MS51521B5H	5/16	Not available	Not available
MS51521A5J	MS51521B5J	5/16	Not available	Not available
MS51521A5M	MS51521B5M	5/16	J514-5-5-070221MA	J514-5-5-070221MB
MS51521A5N	MS51521B5N	5/16	J514-5-5-070221NA	J514-5-5-070221NB
MS51521A5P	MS51521B5P	5/16	J514-5-5-070221PA	J514-5-5-070221PB
MS51521A5R	MS51521B5R	5/16	Not available	Not available
MS51521A5S	MS51521B5S	5/16	J514-5-5-070221SA	J514-5-5-070221SB
MS51521A5T	MS51521B5T	5/16	J514-5-5-070221TA	J514-5-5-070221TB
MS51521A5TF	MS51521B5TF	5/16	Not available	Not available
MS51521A5Z	MS51521B5Z	5/16	J514-5-5-070221ZA	J514-5-5-070221ZB
MS51521A5ZC	MS51521B5ZC	5/16	Not available	Not available
MS51521A6	MS51521B6	3/8	Not available	Not available
MS51521A6H	MS51521B6H	3/8	Not available	Not available
MS51521A6J	MS51521B6J	3/8	Not available	Not available
MS51521A6M	MS51521B6M	3/8	J514-6-6-070221MA	J514-6-6-070221MB
MS51521A6N	MS51521B6N	3/8	J514-6-6-070221NA	J514-6-6-070221NB
MS51521A6P	MS51521B6P	3/8	J514-6-6-070221PA	J514-6-6-070221PB
MS51521A6R	MS51521B6R	3/8	Not available	Not available
MS51521A6S	MS51521B6S	3/8	J514-6-6-070221SA	J514-6-6-070221SB
MS51521A6T	MS51521B6T	3/8	J514-6-6-070221TA	J514-6-6-070221TB
MS51521A6TF	MS51521B6TF	3/8	Not available	Not available
MS51521A6Z	MS51521B6Z	3/8	J514-6-6-070221ZA	J514-6-6-070221ZB
MS51521A6ZC	MS51521B6ZC	3/8	Not available	Not available
MS51521A8	MS51521B8	1/2	Not available	Not available
MS51521A8H	MS51521B8H	1/2	Not available	Not available
MS51521A8J	MS51521B8J	1/2	Not available	Not available
MS51521A8M	MS51521B8M	1/2	J514-8-8-070221MA	J514-8-8-070221MB
MS51521A8N	MS51521B8N	1/2	J514-8-8-070221NA	J514-8-8-070221NB
MS51521A8P	MS51521B8P	1/2	J514-8-8-070221PA	J514-8-8-070221PB
MS51521A8R	MS51521B8R	1/2	Not available	Not available
MS51521A8S	MS51521B8S	1/2	J514-8-8-070221SA	J514-8-8-070221SB
MS51521A8T	MS51521B8T	1/2	J514-8-8-070221TA	J514-8-8-070221TB
MS51521A8TF	MS51521B8TF	1/2	Not available	Not available
MS51521A8Z	MS51521B8Z	1/2	J514-8-8-070221ZA	J514-8-8-070221ZB
MS51521A8ZC	MS51521B8ZC	1/2	Not available	Not available

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TABLE III. MS51521 to SAE-J514 cross reference - Continued.

Inactive for new design MS51518- PIN		Tube O.D.	For new design SAE-J514 PIN	
Assembly	Body		Assembly	Body
MS51521A10	MS51521B10	5/8	Not available	Not available
MS51521A10H	MS51521B10H	5/8	Not available	Not available
MS51521A10J	MS51521B10J	5/8	Not available	Not available
MS51521A10M	MS51521B10M	5/8	J514-10-10-070221MA	J514-10-10-070221MB
MS51521A10N	MS51521B10N	5/8	J514-10-10-070221NA	J514-10-10-070221NB
MS51521A10P	MS51521B10P	5/8	J514-10-10-070221PA	J514-10-10-070221PB
MS51521A10R	MS51521B10R	5/8	Not available	Not available
MS51521A10S	MS51521B10S	5/8	J514-10-10-070221SA	J514-10-10-070221SB
MS51521A10T	MS51521B10T	5/8	J514-10-10-070221TA	J514-10-10-070221TB
MS51521A10TF	MS51521B10TF	5/8	Not available	Not available
MS51521A10Z	MS51521B10Z	5/8	J514-10-10-070221ZA	J514-10-10-070221ZB
MS51521A10ZC	MS51521B10ZC	5/8	Not available	Not available
MS51521A12	MS51521B12	3/4	Not available	Not available
MS51521A12H	MS51521B12H	3/4	Not available	Not available
MS51521A12J	MS51521B12J	3/4	Not available	Not available
MS51521A12M	MS51521B12M	3/4	J514-12-12-070221MA	J514-12-12-070221MB
MS51521A12N	MS51521B12N	3/4	J514-12-12-070221NA	J514-12-12-070221NB
MS51521A12P	MS51521B12P	3/4	J514-12-12-070221PA	J514-12-12-070221PB
MS51521A12R	MS51521B12R	3/4	Not available	Not available
MS51521A12S	MS51521B12S	3/4	J514-12-12-070221SA	J514-12-12-070221SB
MS51521A12T	MS51521B12T	3/4	J514-12-12-070221TA	J514-12-12-070221TB
MS51521A12TF	MS51521B12TF	3/4	Not available	Not available
MS51521A12Z	MS51521B12Z	3/4	J514-12-12-070221ZA	J514-12-12-070221ZB
MS51521A12ZC	MS51521B12ZC	3/4	Not available	Not available
MS51521A14	MS51521B14	7/8	Not available	Not available
MS51521A14H	MS51521B14H	7/8	Not available	Not available
MS51521A14J	MS51521B14J	7/8	Not available	Not available
MS51521A14M	MS51521B14M	7/8	J5141-4-14-070221MA	J5141-4-14-070221MB
MS51521A14N	MS51521B14N	7/8	J5141-4-14-070221NA	J5141-4-14-070221NB
MS51521A14P	MS51521B14P	7/8	J5141-4-14-070221PA	J5141-4-14-070221PB
MS51521A14R	MS51521B14R	7/8	Not available	Not available
MS51521A14S	MS51521B14S	7/8	J5141-4-14-070221SA	J5141-4-14-070221SB
MS51521A14T	MS51521B14T	7/8	J5141-4-14-070221TA	J5141-4-14-070221TB
MS51521A14TF	MS51521B14TF	7/8	Not available	Not available
MS51521A14Z	MS51521B14Z	7/8	J5141-4-14-070221ZA	J5141-4-14-070221ZB
MS51521A14ZC	MS51521B14ZC	7/8	Not available	Not available

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TABLE III. MS51521 to SAE-J514 cross reference - Continued.

Inactive for new design MS51518- PIN		Tube O.D.	For new design SAE-J514 PIN	
Assembly	Body		Assembly	Body
MS51521A16	MS51521B16	1	Not available	Not available
MS51521A16H	MS51521B16H	1	Not available	Not available
MS51521A16J	MS51521B16J	1	Not available	Not available
MS51521A16M	MS51521B16M	1	J5141-6-16-070221MA	J5141-6-16-070221MB
MS51521A16N	MS51521B16N	1	J5141-6-16-070221NA	J5141-6-16-070221NB
MS51521A16P	MS51521B16P	1	J5141-6-16-070221PA	J5141-6-16-070221PB
MS51521A16R	MS51521B16R	1	Not available	Not available
MS51521A16S	MS51521B16S	1	J5141-6-16-070221SA	J5141-6-16-070221SB
MS51521A16T	MS51521B16T	1	J5141-6-16-070221TA	J5141-6-16-070221TB
MS51521A16TF	MS51521B16TF	1	Not available	Not available
MS51521A16Z	MS51521B16Z	1	J5141-6-16-070221ZA	J5141-6-16-070221ZB
MS51521A16ZC	MS51521B16ZC	1	Not available	Not available
MS51521A20	MS51521B20	1 1/4	Not available	Not available
MS51521A20H	MS51521B20H	1 1/4	Not available	Not available
MS51521A20J	MS51521B20J	1 1/4	Not available	Not available
MS51521A20M	MS51521B20M	1 1/4	J514-20-20-070221MA	J514-20-20-070221MB
MS51521A20N	MS51521B20N	1 1/4	J514-20-20-070221NA	J514-20-20-070221NB
MS51521A20P	MS51521B20P	1 1/4	J514-20-20-070221PA	J514-20-20-070221PB
MS51521A20R	MS51521B20R	1 1/4	Not available	Not available
MS51521A20S	MS51521B20S	1 1/4	J514-20-20-070221SA	J514-20-20-070221SB
MS51521A20T	MS51521B20T	1 1/4	J514-20-20-070221TA	J514-20-20-070221TB
MS51521A20TF	MS51521B20TF	1 1/4	Not available	Not available
MS51521A20Z	MS51521B20Z	1 1/4	J514-20-20-070221ZA	J514-20-20-070221ZB
MS51521A20ZC	MS51521B20ZC	1 1/4	Not available	Not available
MS51521A24	MS51521B24	1 1/2	Not available	Not available
MS51521A24G	MS51521B24G	1 1/2	Not available	Not available
MS51521A24H	MS51521B24H	1 1/2	Not available	Not available
MS51521A24M	MS51521B24M	1 1/2	J514-24-24-070221MA	J514-24-24-070221MB
MS51521A24N	MS51521B24N	1 1/2	J514-24-24-070221NA	J514-24-24-070221MB
MS51521A24P	MS51521B24P	1 1/2	J514-24-24-070221PA	J514-24-24-070221PB
MS51521A24R	MS51521B24R	1 1/2	Not available	Not available
MS51521A24S	MS51521B24S	1 1/2	J514-24-24-070221SA	J514-24-24-070221SB
MS51521A24T	MS51521B24T	1 1/2	J514-24-24-070221TA	J514-24-24-070221TB
MS51521A24TF	MS51521B24TF	1 1/2	Not available	Not available
MS51521A24Z	MS51521B24Z	1 1/2	J514-24-24-070221ZA	J514-24-24-070221ZB
MS51521A24ZC	MS51521B24ZC	1 1/2	Not available	Not available

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TABLE III. MS51521 to SAE-J514 cross reference - Continued.

Inactive for new design MS51518- PIN		Tube O.D.	For new design SAE-J514 PIN	
Assembly	Body		Assembly	Body
MS51521A32	MS51521B32	2	Not available	Not available
MS51521A32H	MS51521B32H	2	Not available	Not available
MS51521A32J	MS51521B32J	2	Not available	Not available
MS51521A32M	MS51521B32M	2	J514-32-32-070221MA	J514-32-32-070221MB
MS51521A32N	MS51521B32N	2	J514-32-32-070221NA	J514-32-32-070221NB
MS51521A32P	MS51521B32P	2	J514-32-32-070221PA	J514-32-32-070221PB
MS51521A32R	MS51521B32R	2	Not available	Not available
MS51521A32S	MS51521B32S	2	J514-32-32-070221SA	J514-32-32-070221SB
MS51521A32T	MS51521B32T	2	J514-32-32-070221TA	J514-32-32-070221TB
MS51521A32TF	MS51521B32TF	2	Not available	Not available
MS51521A32Z	MS51521B32ZZ	2	J514-32-32-070221ZA	J514-32-32-070221ZB
MS51521A32ZC	MS51521B32ZC	2	Not available	Not available

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-18866, this document references the following:

FED-STD-595/36076	FED-STD-595/36280	ASTM B166	SAE-AMS2700
FED-STD-595/36081	FED-STD-595/36293	ASTM B564	SAE-AMS4928
FED-STD-595/36099	MIL-DTL-16232	ASTM B633	SAE-AMS5639
FED-STD-595/36118	MS51531	ASTM B695	SAE-AMS5645
FED-STD-595/36134	MS51533	ASTM F1136/F1136M	SAE-AMS5647
FED-STD-595/36152	QQ-N-281	SAE-AMS-C-81562	SAE-AMS5743
FED-STD-595/36170	ASME B1.1	SAE-AMS-QQ-P-416	SAE-AMS6370
FED-STD-595/36173	ASTM A276/A276M	SAE-AMS2417	SAE-AMS6382
FED-STD-595/36176	ASTM A564/A564M	SAE-AMS2451/5	SAE-J403
FED-STD-595/36231	ASTM A582/A582M	SAE-AMS2451/9	SAE-J425
FED-STD-595/36251	ASTM B117	SAE-AMS-2486	SAE-J514
FED-STD-595/36270	ASTM B164	SAE-AMS2488	

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CONCLUDING MATERIAL

Custodians:
Army - AR
Navy - OS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2016-021)

Review activities:
Army - AT, MI
Navy - CG, MC, SA, SH
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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